Data Booklet

Year 11 Psychology

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How to add data toolpak to your laptop

1. Click the **Tools** menu, and then click **Excel Add-ins**.

2. In the **Add-Ins available** box, select the **Analysis ToolPak** check box, and then click **OK**.

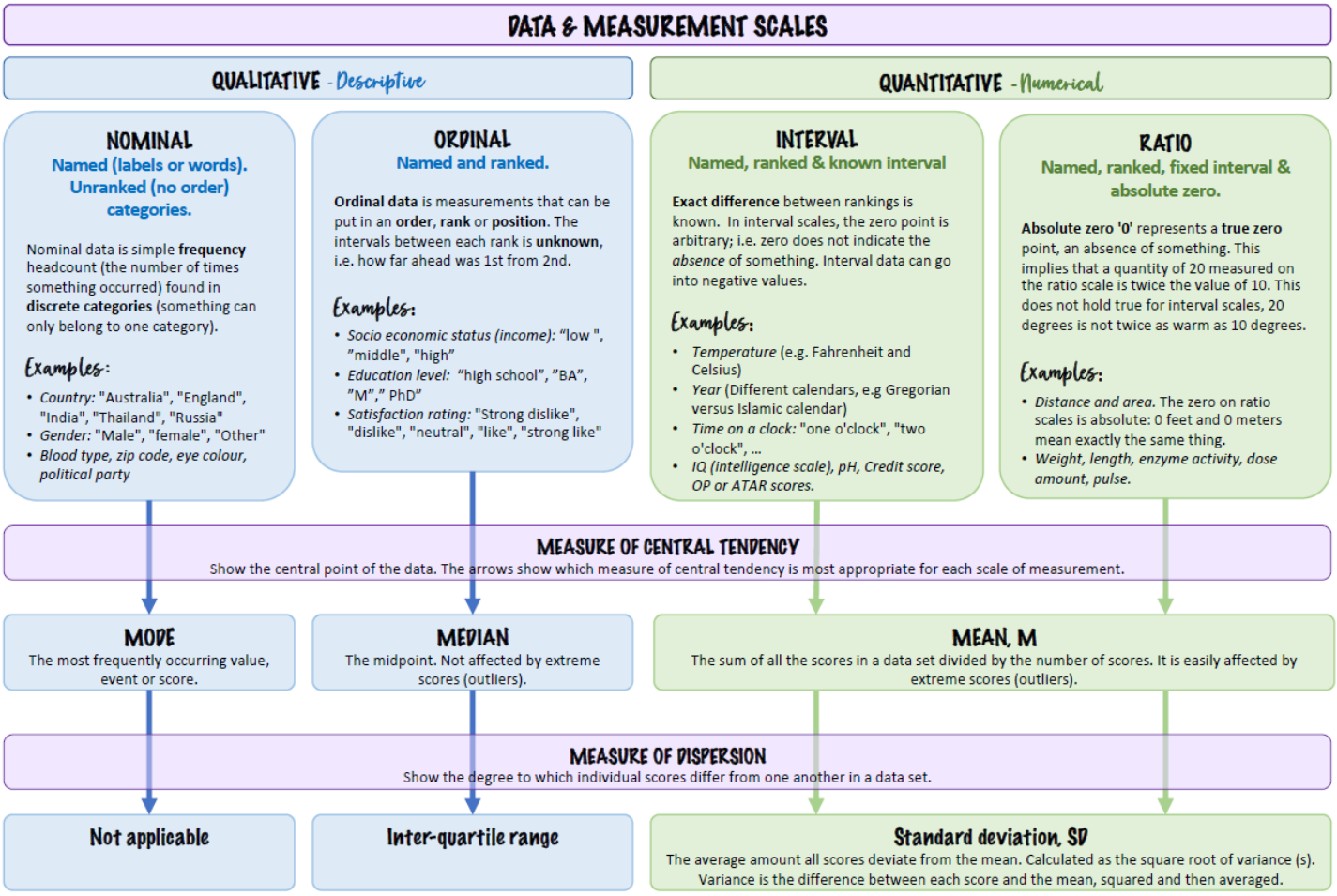
   a. If **Analysis ToolPak** is not listed in the **Add-Ins available** box, click **Browse** to locate it.

   b. If you get a prompt that the Analysis ToolPak is not currently installed on your computer, click **Yes** to install it.

   c. Quit and restart Excel.

   Now the **Data Analysis** command is available on the **Data** tab.

Data collection



https://www.youtube.com/watch?v=rL38g06DbSc video on the 4 types of data

Subjective and objective data

Subjective data is information about the variables being studied based on opinion, with no external yardstick by which they are measured. (e.g., feelings)

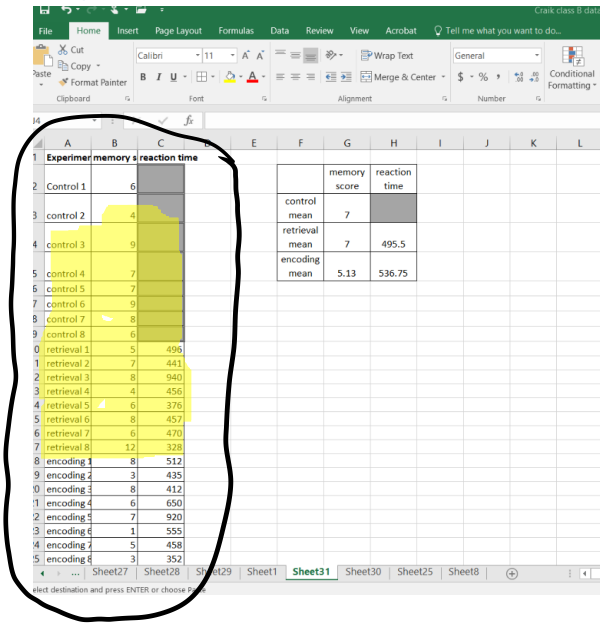
Objective data is data that is measured according to an identifiable external criterion. (e.g., standardized tests)

Descriptive statistics

How to work out descriptive stats in toolpak

Step 1: *Add your raw data to a table (see image 1 below).*

Image 1: Table of raw data



Step 2: *You can then use the data toolpak to help you work out the descriptive statistics. You will need the means to create a column or line graph.*

To work out the descriptive statistics:

Step 3: *Click on Data then, data analysis and then descriptive statistics. Ensure you tick summary statistics. (See image 2 and 3 below)*

Image 2: Using toolpak

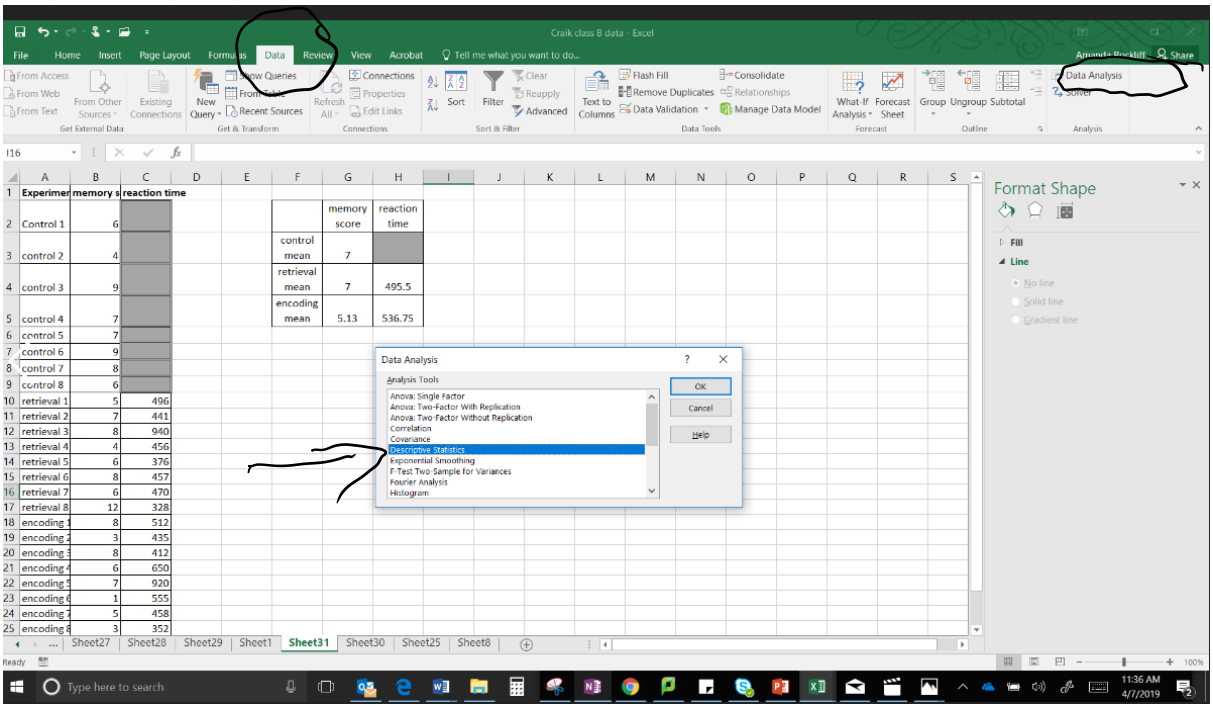
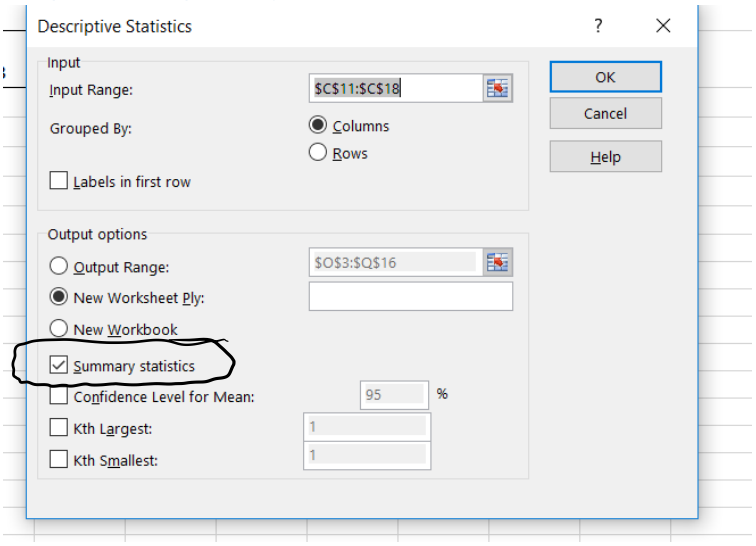


Image 3: Adding descriptive stats



Step 4: *Input range is the data you wish to input. For the raw data above, you will have 5 sets of data to enter. You do each of these data sets separately. Make sure you clearly label the data you receive so you can find it easily.*

For example: I chose the first set of data (highlighted in Image 1). I selected the numbers only in the input range in Image 3. I checked that summary statistics was ticked then chose my output range as a new worksheet Ply. Then I pressed ok.

This is the information I got:

Image 4: descriptive stats output.

control group memory	
Mean	7
Standard E	0.597614
Median	7
Mode	6
Standard D	1.690309
Sample Va	2.857143
Kurtosis	-0.028
Skewness	-0.47329
Range	5
Minimum	4
Maximum	9
Sum	56
Count	8

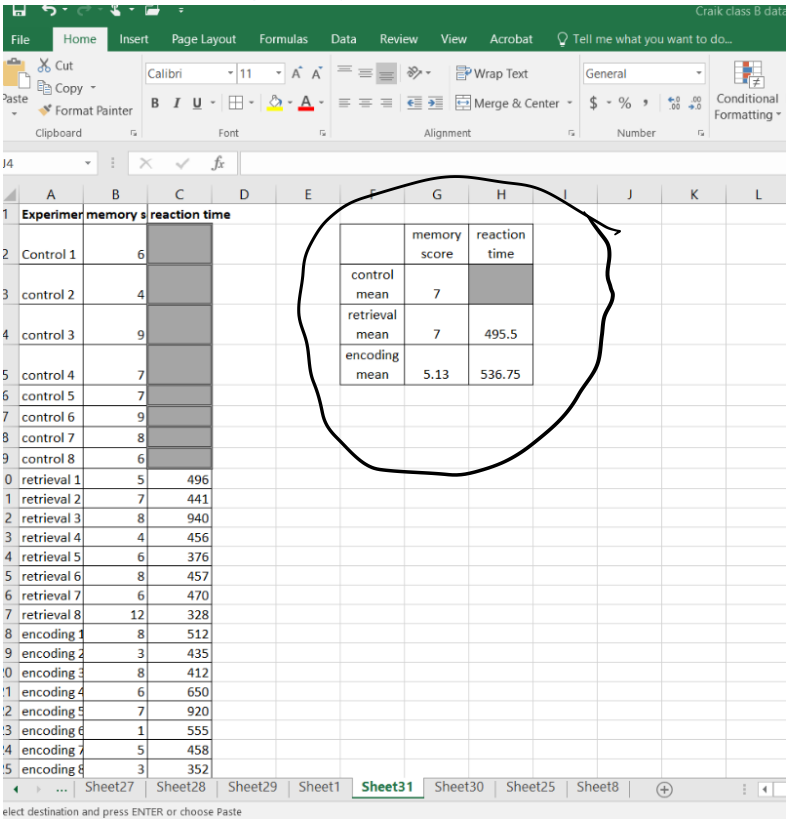
Step 5: *Change the title so that you remember what this set of data was for.*

You now have the measures of central tendency so you can begin displaying data in visual form.

How to display data in excel

Create a small table in excel with the information from your descriptive stats (see image 5)

Image 5: summarizing descriptive statistics



You can then use this information to help you create graphs.

Step 6: *To create a graph, select the data you wish to graph. (e.g., the 3 mean memory scores) and then at the top of your screen in excel, choose Insert and then click on your desired chart type (see image 6)*

Image 6: Inserting charts

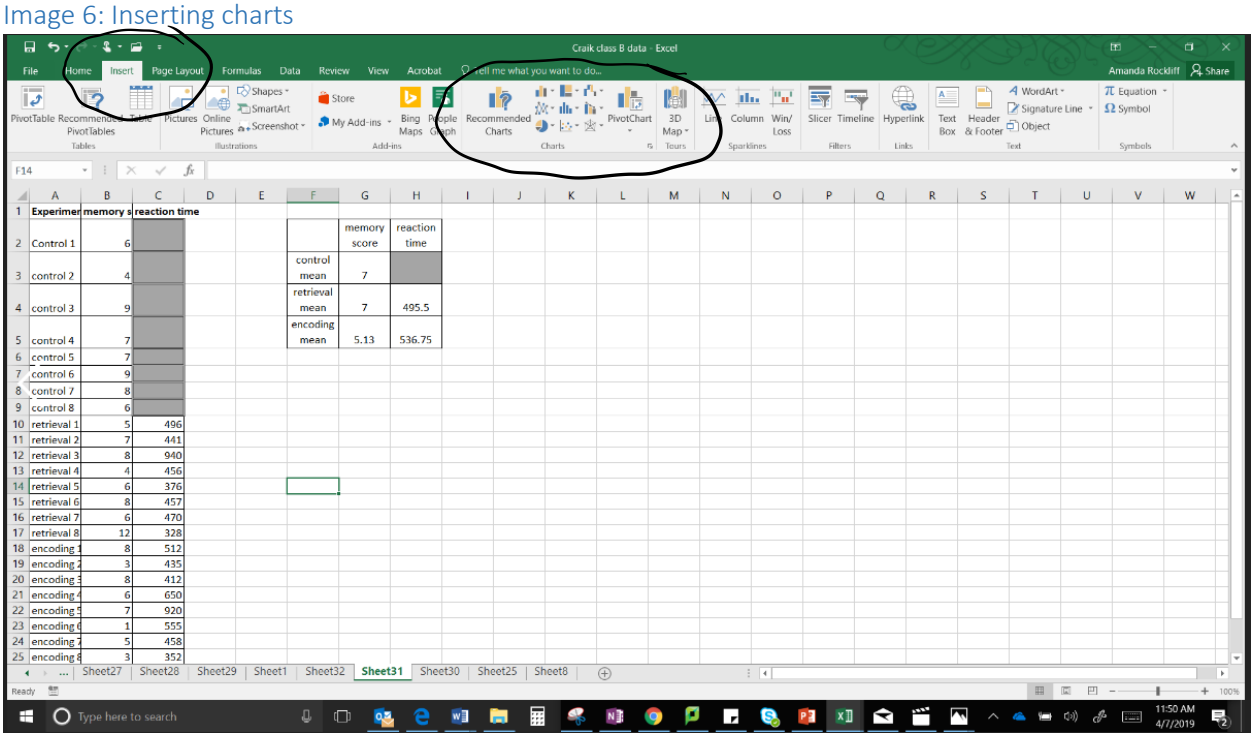
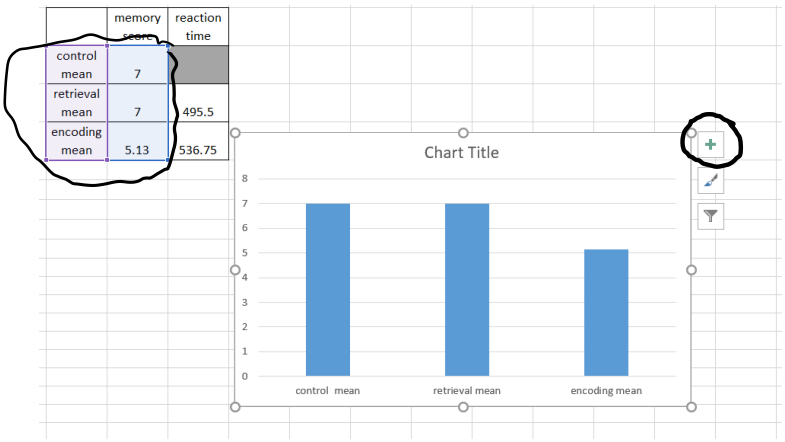


Image 7: chart of memory means



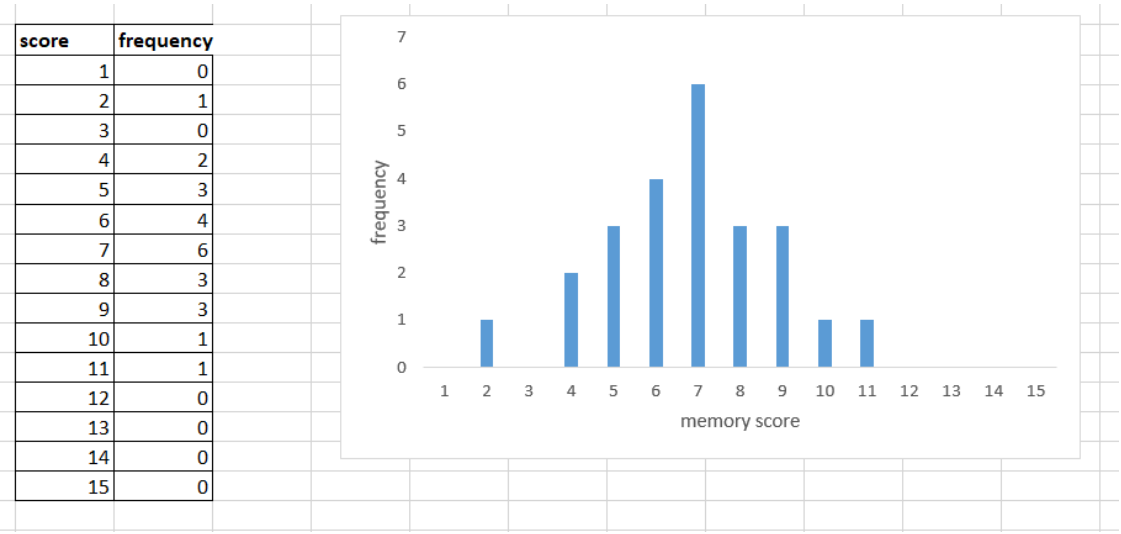
Step 7: *You can now use the small plus button on the side of the chart to add in features such as:*

- axis titles
- removing gridlines
- changing colour (plain black) – see screenshots on confidence intervals to see how to do this.
- Remove chart title (chart titles should be below the chart)

You can add a variety of different charts (see the section on correlations to see how to add scatterplots)

Step 8: *To plot your data on a frequency graph, create a frequency table in excel (see below) and then add a frequency chart in excel. See below (Image 9) for an example:*

Image 8: frequency table and chart

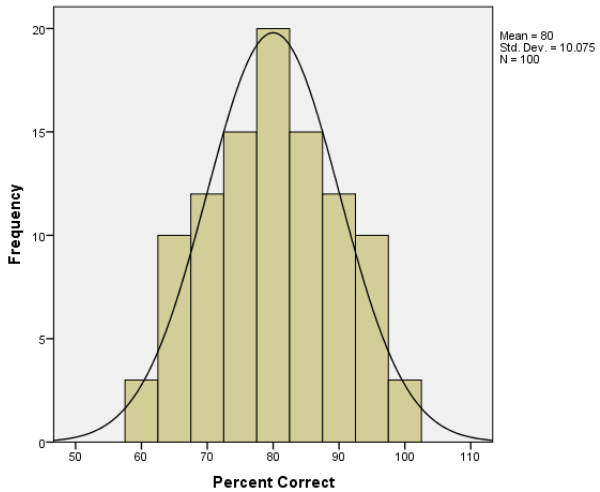


score	frequency
1	0
2	1
3	0
4	2
5	3
6	4
7	6
8	3
9	3
10	1
11	1
12	0
13	0
14	0
15	0

How to determine if your data fits the normal curve:

- Ideally results from an experiment will follow a **normal distribution**
- Results from an experiment with normal distribution will have most of their data around a central value.
- Distribution graphs are created by calculating the frequency a data point occurs and plotting this on a graph. (Frequency charts (see Image 9) can be added in excel to help you determine your distribution)

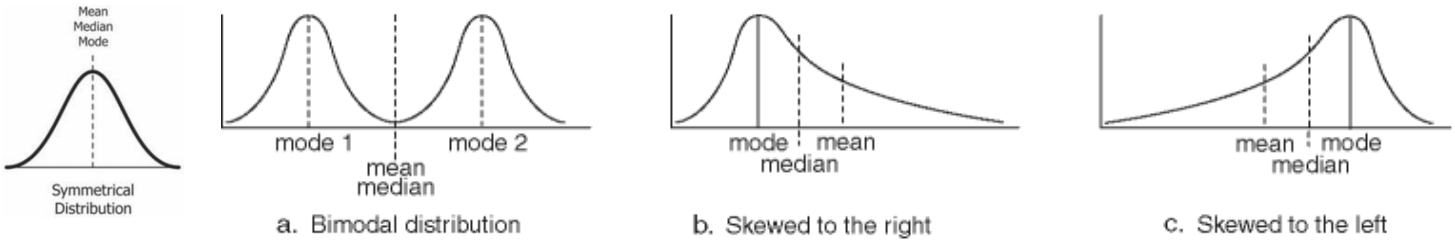
Image 9: Frequency chart with a bell curve (normal distribution)



Skew of a curve

It is rare for all the data in a dataset to be able to be plotted onto a smooth, well-balanced curve like the one in image 9. More often the plot is biased or skewed one way or the other (positive or negative). Sometimes it will have more than one peak- this is called a bimodal distribution.

Image 10: skew of a curve



Measures of central tendency - mean ($\bar{x}$), median and mode

Measures of central tendency tell us how the data are clustered near the central point of the dataset. For the normal curve, the three measures mean, median and mode are all at the same point. Central tendency refers to the idea that there is one number that **best summarizes** the data set, a number that is in some way "central" to the set

Mean

It is the average of all the scores (add all scores and divide by the number of scores)

- $\bar{x}$ = sample mean
- μ = population mean
- n = sample size
- N = population size (note that many QCAA tasks have used a capital N to refer to sample size)

Formula to use:

Use the formula $\bar{x} = \frac{x_1+x_2+\cdots+x_N}{N}$ where x =raw data point and N= sample size.

Round your answer to the nearest whole number.

Sample Mean	Population Mean
$\bar{x} = \frac{\Sigma x}{n}$	$\mu = \frac{\Sigma x}{N}$

where Σx is sum of all data values

N is number of data items in population

n is number of data items in sample

Median

It is the middle number (or mean of the two middle numbers) of a number series. Divides the data in 2 equal parts i.e., number of terms on right side of it is same as number of terms on left side of it when data is arranged in either ascending or descending order.

- If no central number than the average of the 2 middle values

  5, 7, 9, 9,[11], 11, 15, 17, 19

  or

  5, 7, 9, 9, [11, 13], 15, 17, 17, 19

  $\frac{11+13}{2}$ **= 12**

Mode

It is the most commonly occurring score. The term appearing maximum time in data set i.e., term that has highest frequency.

12, 24, 41, 51, 67, 67, 85, 99

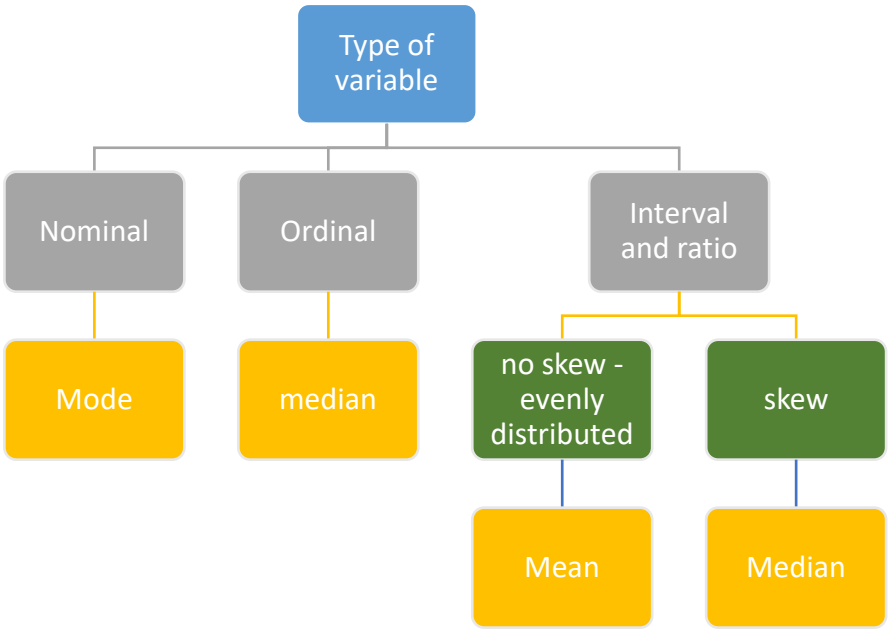
Some data sets may have no mode at all as all values appears same number of times.

- If two values appeared same time and more than the rest of the values, then the data set is **bimodal**.
- If three values appeared same time and more than the rest of the values then the data set is **trimodal** and for n modes, that data set is **multimodal**

How to choose the best measure of central tendency

This depends on outliers, and data not fitting a normal curve (See Table 1).

Table 1: Best measures of central tendency



Measures of variability (dispersion)

These tell us how the data is spread out. Together with the measures of central tendency they tell us about the features of the dataset. They aim to provide an indication of how well the mean represents the data.

Large spread indicates large differences in data/individual scores. These include:

Range

The difference between the highest score and the lowest score in the dataset.

Variance

A measure of how much, on average, the scores differ from the mean. This is not as useful as the standard deviation.

It is a more useful measure than range as it compares each data point to the mean to calculate on average how much the scores differ from the mean.

- It assumes **normal distribution**** of data

BUT!

- Some values are lower than the mean and some are above?? Thus, if you did this the negative and the positive values would 'equal' out and provide you with an inaccurate score!
- To overcome this all differences are squared – so that all differences are positive!

To calculate variance

- Calculate the mean $\bar{x}$
- Calculate the difference between each data point (x_i) and the mean ($\bar{x}$) then square this difference $(x_i - \bar{x})^2$
- Calculate mean of the 'squared differences'

*Remember squared means to multiply a number by itself.

Image 11: Calculating variance - Distribution in a normal curve

Example: Calculating variance

Data 1
5
5.4
3
4
4.2
6

1. Calculate the sample mean $\bar{x}$ = 27.6/6= 4.6
2. Calculate each scores difference from the mean and square it.

Data 1	Difference from mean	Square
5	0.4	0.16
5.4	0.8	0.64
3	-1.6	2.56
4	-0.6	0.36
4.2	-0.4	0.16
6	1.4	1.96

3. Calculate mean of square = 5.84/6 = 0.97

Therefore the variance is 0.97

a small variance indicates highly similar scores, all close to the sample mean, and a large variance indicates more scores at a greater distance from the mean and possibly spread over a larger range

- However, the variance doesn't tell us much about the spread of data as its in squared units.
- Therefore, we need to go one step further and calculate the sample standard deviation (s)

Standard deviation (S)

Determines how far on average, scores are different from the mean.

Used for (population data) OR standard error (sample data).

- Standard deviation is **calculated** by taking the square root of the variance. (Can be automatically done on data toolpak in excel)
- If we look at the data set from above the variance was 0.97
- Standard deviation of the sample (data 1) ($SD\ \bar{x}$) = $\sqrt{0.97}$ = 0.98

Interpreting Standard Deviation

- A low standard deviation means that most of the numbers are very close to the mean.
- A high standard deviation means that the numbers are spread out therefore will have a higher variation!

- Our standard deviation was 0.98: Therefore approx. 68.26% of values sit within 1 standard deviation from the mean (See Image 12)

Image 12: Standard Deviation – Distribution in a normal curve

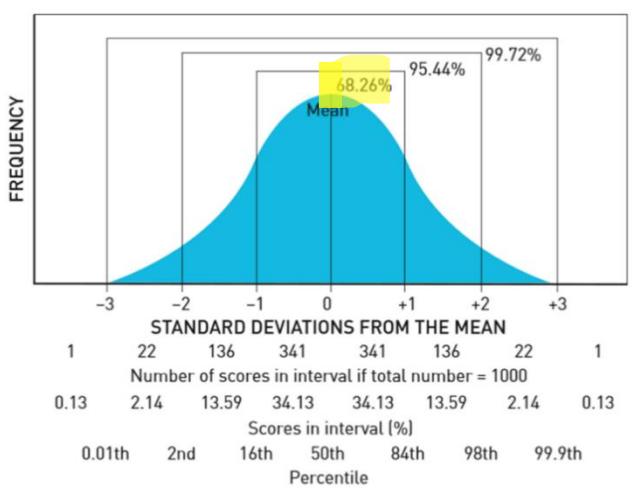


FIGURE 8 Distribution in a normal curve, showing percentiles by standard deviation

Standard error (of the mean) (SE_{μ_x})

- The standard error takes the standard deviation from the sample and generalises it for the whole population.
- Standard error theoretically should equal the standard deviation of the population
- It is difficult to calculate the standard deviation of an entire population (think about it)
- Standard error of the mean (SE) measures how accurately the sample mean ($\bar{x}$) represents the average population mean (μ).

Interpreting Standard error (of the mean)

- A **small SE** indicates that the sample mean ($\bar{x}$) is an accurate reflection of the actual population mean μ.
- A **large SE** makes us **less confident** that the sample mean ($\bar{x}$) represents the population mean (μ).

$$SE_{\mu_x} = \frac{s}{\sqrt{n}}$$

SE_{μ_x} = Standard error of the mean

S = Standard deviation

n = total number of data sets

Percentiles

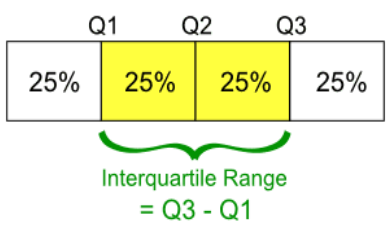
- Percentile is a way to represent position of a value in a data set.
- To calculate percentile, values in data set should always be in ascending order.
  12, 24, 41, 51, 67, 67, 85, 99

Interpreting percentiles

- The median 59 has 4 values less than itself out of 8.
- It can also be said as: In the data set, 59 is the 50th percentile because 50% of the total terms are less than 59.

Quartiles

- The lower quartile (Q1) is the point between the lowest 25% of values and the highest 75% of values. It is also called the 25th percentile.
- The second quartile (Q2) is the middle of the data set. It is also called the 50th percentile, or the median.
- The upper quartile (Q3) is the point between the lowest 75% and highest 25% of values. It is also called the 75th percentile



Interpreting quartiles

For the following data set:

12, 24, 41, 51, 67, 67, 85, 99, 115

Q2 = 67: is at the 50th percentile of the whole data and is median.

Q1 = 41: is at the 25th percentile of the data.

Q3 = 85: is at the 75th percentile of the date.

Interquartile range (IQR) = Q3 - Q1 = 85 - 41 = 44

Adding descriptive statistics to graphs

- When graphing data, it is important to provide an indication of the variation in your data set.
- These are presented as error bars.

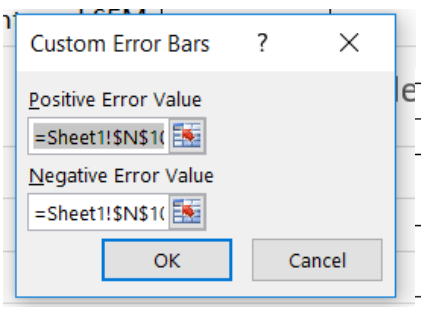
Error bars are (often) the **standard error** (can also be standard deviation)

- **Psychology texts generally will show standard error**

Adding error bars in excel

Please see below for the instructions on adding confidence levels. <u>Repeat this exact same procedure for adding error bars,</u> except- when you choose the data for your custom error bars, you select the standard error of the mean cells.

Image 13: custom error bars



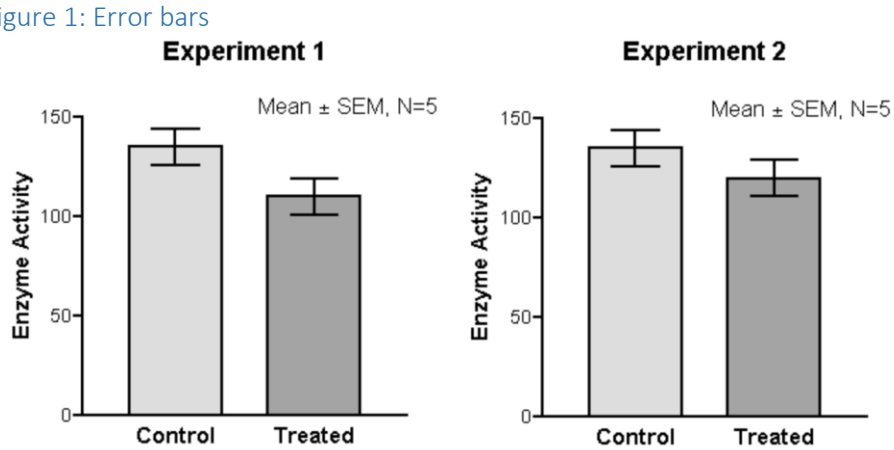
	control	retrieval	encoding
Memory mean	7	7	5.13
standard error of the mean	0.60	0.87	0.91
confidence interval SEM x 1.96	1.17	1.70	1.79

Interpreting standard error bars

When standard error (SE) bars **do not overlap**, you **cannot be sure** that the difference between two means is statistically significant. (Therefore, might be statistically significant!)

When SE bars overlap, (as in experiment 2) you can be sure the difference between the two means is **not statistically significant**

Figure 1: Error bars



See confidence intervals for how to change the colour and remove columns.

Confidence intervals (CI)

- SD (standard deviation) tells you on **average** the variation your sample data is from your sample mean
- SE (standard error) tells you on ***average*** the variation from the population mean
- Confidence intervals (CI) tell you the likelihood the **population mean** will fit in between two values (the upper and lower confidence interval)

For example.

You are 95% confident that the population mean fits between 10.5 seconds – 15 seconds.

Calculates how likely (%) your **sample mean** $\overline{x}$ represents your **population mean μ.**

For example: *How confident are we that the sample mean response time of our sample is representative of a whole population?*

How to work out confidence intervals:

First, we need to specify our confidence interval

- Common confidence intervals
  - *We are 90% confident*
  - *We are 95% confident*
  - *We are 99% confident*
- The more confident we wish to be the larger our confidence interval will be!
- **Generally, a 95% confident interval is used.**

To calculate the confidence interval:

Confidence interval = standard error of the mean x 1.96

CI = $SE_{\mu_x}\ x\ 1.96$

This can be done in excel with the following formula:

Click on the empty cell and type = then click on the standard error cell (it will appear in the empty cell as a letter and number), then type *1.96 press enter and it will calculate. You can also use your calculator.

Image 14: raw data

Column1		
Mean	2.68	
Standard Error	0.262805	x 1.96= confidence interval
Median	2	
Mode	2	
Standard Deviation	1.314027	
Sample Variance	1.726667	
Kurtosis	-0.89	
Skewness	0.411234	
Range	4	
Minimum	1	
Maximum	5	
Sum	67	
Count	25	

Image 15: Table of central tendencies

	control	retrieval	encoding
Memory mean	7	7	5.13
standard error of the mean	0.60	0.87	0.91
confidence interval SEM x 1.96	1.17	1.70	1.79

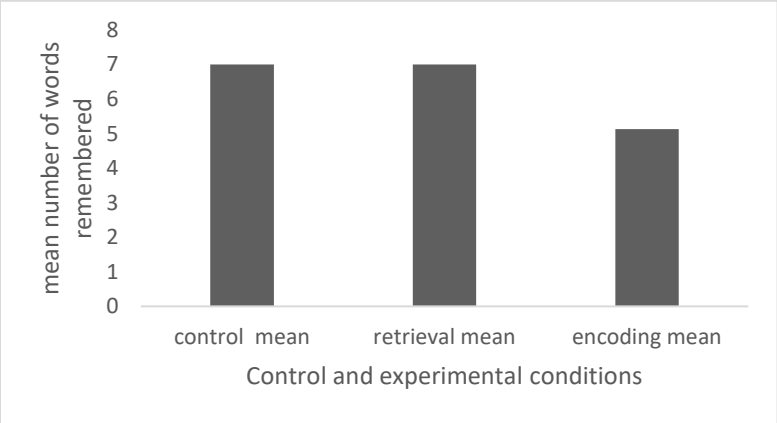
	control	retrieval	encoding
reaction mean		495.50	536.75
standard error of the mean		66.36	63.65
confidence interval SEM x 1.96		130.06	124.76

Create a table with this data (See Image 15).

How to add Standard error of the mean or confidence intervals in excel

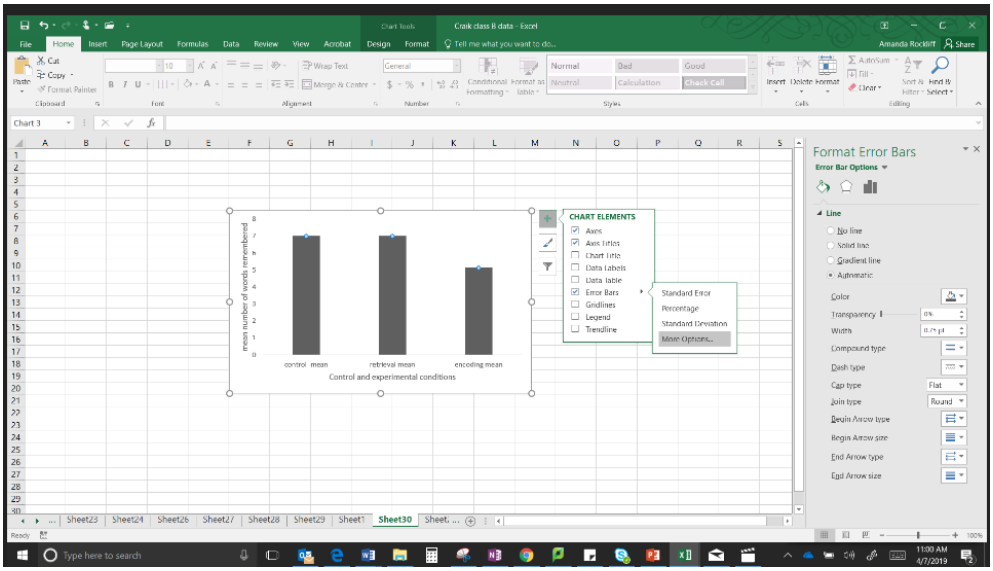
With the data above I had the following graph for memory:

Figure 2: Mean scores for number of words recalled in control and experimental conditions.



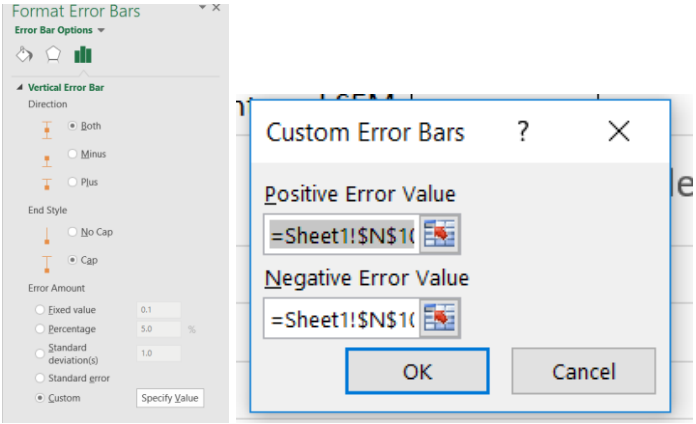
Click on the + and select error bars and more options (See Image 16)

Image 16: adding confidence/ SEM intervals



Then choose custom and specify value. The box below will pop up- custom error bars.

Image 17: custom error bars



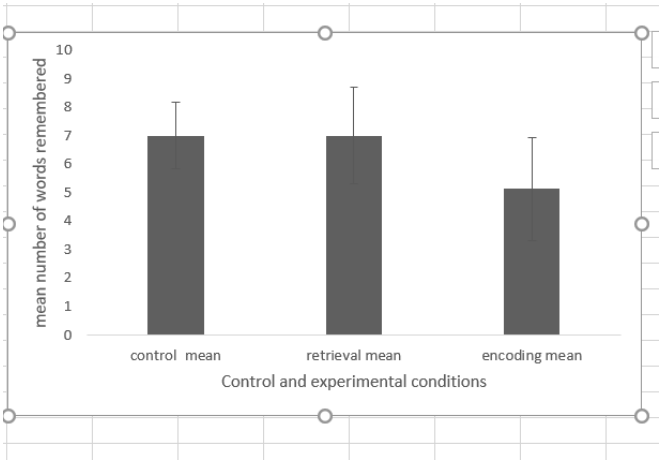
In the first cell, copy the confidence or SEM levels , then repeat this in the second cell and press ok.

Image 18: confidence interval data

	control	retrieval	encoding
Memory mean	7	7	5.13
standard error of the mean	0.60	0.87	0.91
confidence interval SEM x 1.96	1.17	1.70	1.79

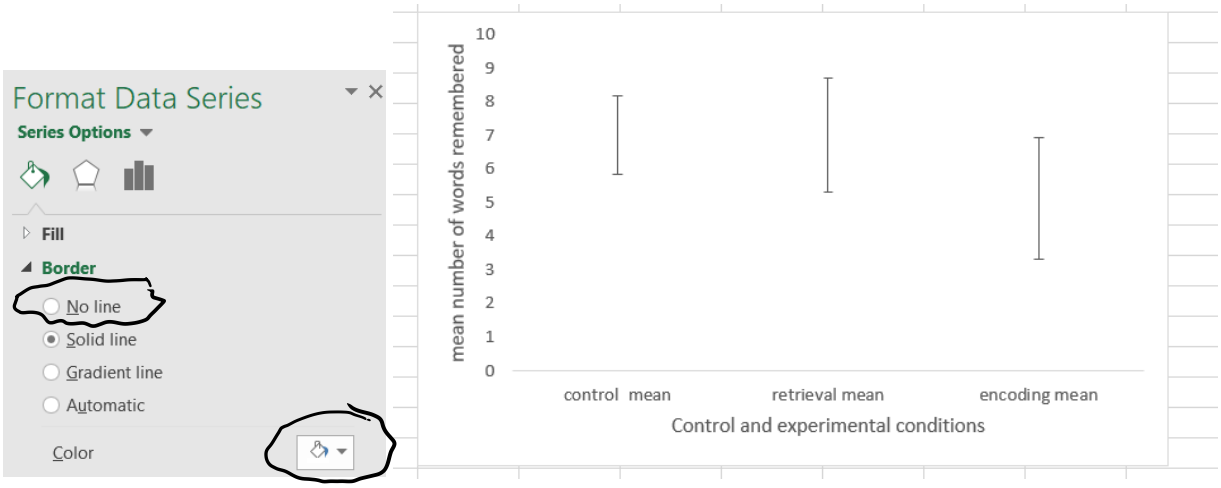
When I press okay, my graph now looked like this:

Image 19: confidence intervals/ SEM on a column graph



If I want to put the focus on the error bars- to see if they overlap or not, I can remove the columns by making them white, using format data series.

Image 20: removing columns from confidence intervals and error bars



Excel formulas for descriptive statistics

Descriptive statistic	Excel formula
Mean	=AVERAGE
Median	=MEDIAN
Mode	=MODE
Range	=MIN & =MAX
Quartiles	=QUARTILE (array, quart) (array is your data, quart is the number of quartiles generally use D4)
Skewness	=SKEW (close to 0 = more normal distribution)
Variance	=VAR (sample) =VARP (population)
Standard deviation	=STDEV.P (population) =STDEV.S (sample)
Standard error	=STDEV(''cell range'')/SQRT(count("cell range"))
Confidence intervals	=CONFIDENCE(alpha,sigma,n) Alpha is the confidence interval gen. 0.05 Sigma = standard deviation of sample N=sample size

Inferential statistics

- After the experiment you will need to decide if the results could be due to chance alone (therefore no cause-and-effect relationship can be inferred).
- To do this – calculate mean score for the E-group (experimental) and the C-group (control) and use **inferential statistics** to decide if the difference is statistically significant or is most likely due to chance.
- While the means are different, are they different enough to show that the IV has caused significant change?
- This is where you need to run an inferential test.

P-values

p-value refers to the probability that any difference is due to chance.

If $p \leq 0.05$ THEN you can accept that an observed difference is statistically significant. This means that a difference is statistically significant if the difference could have been caused by chance of 5 or less times in 100.

e.g. If *p=0.03,* then there are 3 chances in 100 (3%) that the difference is caused by chance.

Image 21: P-value example using a pizza business

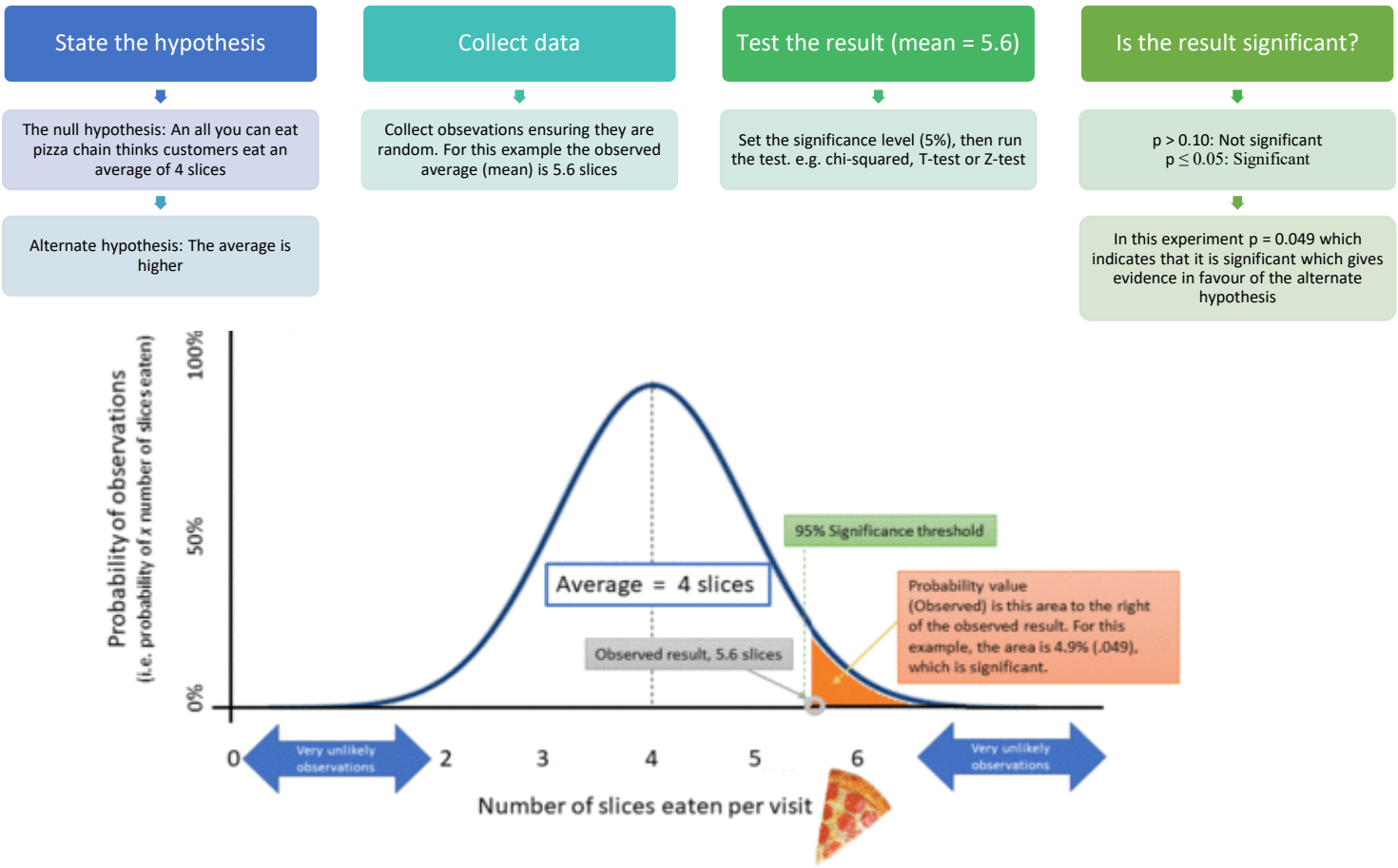


Image 22: Interpreting p-values

P ≥ 0.1	Absence of evidence against the null hypothesis: data consistent with the null hypothesis
0.05 ≤ P < 0.1	Low evidence against the null hypothesis in favour of the alternative
0.01 ≤ P < 0.05	Moderate evidence against the null hypothesis in favour of the alternative
0.001 ≤ P < 0.01	Strong evidence against the null hypothesis in favour of the alternative
P < 0.001	Very strong evidence against the null hypothesis in favour of the alternative

- A **small** or **low** P value (usually below 5% or ≤0.05) suggests that your null hypothesis may be **incorrect**, giving evidence for the alternative hypothesis. This allows us to determine that an effect is present and is applicable to generalise conclusions to a population.
  If the p-value ,0.05 then the results are due to more then chance/coincidence. You can then conclude that there is a cause-effect relationship.
- A **large** P value suggests that your null hypothesis is **correct**

Choosing tests to determine p values and significance

Image 23: Choosing parametric tests

CHOOSING A STATISTICAL TEST

Q1 Type of design?
Are you testing differences between means (experimental) or relationship between variables (correlational)? If experimental, what type of design?

Q2: What type of data do you have?
(refer to "data and measurement scales for help)

Q3: Parametric conditions
1. Interval or ratio data
2. Normal distribution or normal population (n > 30)
3. Homogeneity of variance *(not tested at high school)*

	Q1 Type of design		
	Difference between means Type of experimental design?		Relationship between two variables
Q2. Type of data / levels of measure	**Independent** measures design	**Dependent** *(repeated measures or matched pairs)* design	**Correlation**
NOMINAL DATA – has no meaningful order e.g. gender	Chi squared test	Sign test	Chi squared test
ORDINAL DATA – has an obvious order e.g. Likert scale	Mann Whitney U test	Wilcoxon signed ranks test	Spearman's rho correlation
INTERVAL OR RATIO DATA **Parametric** tests = most powerful and sensitive tests.	Independent *(unpaired)* T-test	Repeated *(paired)* T-test	Pearson's product moment correlation coefficient

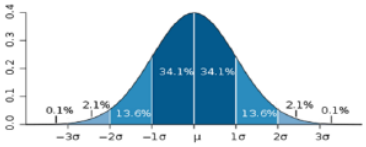
ype I Error - False Positive: *Finding a "significant effect" (p < 0.05) when there is no true ffect. So saying something is wrong when it is in fact right.*

ype II Error - False Negative: *Failing to find a "significant effect" (p < 0.05) when there is a rue effect.*

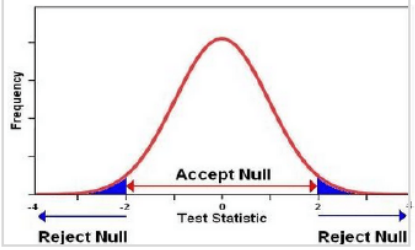
MEMORY AIDE for the table:
I'm Really Cool
No Cars Should Crash
Only Men Wear Socks
Injured Unicorns Roar Painfully

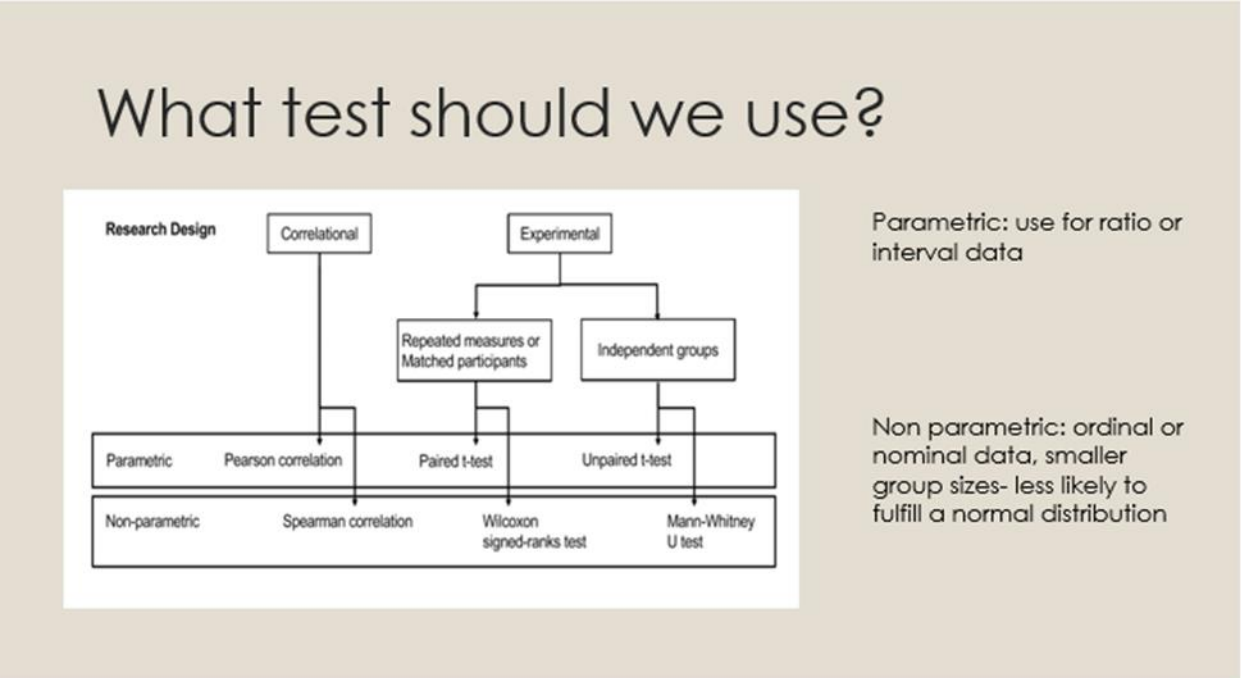
Statistical significance *– the **probability** (p-value) of the observed effect being due to chance (i.e. not a true difference).*

***p*-values:**
- *p*-value refers to the probability that any difference is due to chance.
- **IF *p* ≤ 0.05 THEN** you can accept that an observed difference is statistically significant.



If using Observed and Critical values:
- The ***Mann Whitney*** and ***Wilcoxon tests*** require observed values to be **equal to or LESS** than the critical value to be accepted as significant, allowing the rejection of the null hypothesis. (i.e. Obs Val ≤ Crit Val)
- WHILE on the other hand the **Chi squared and t-tests** require the observed value be **equal to or GREATER** than the critical value to be accepted as significant. (i.e. Obs Val ≥ Crit Val)





How do I know if I should choose a parametric or a non-parametric test?

Parametric assumes normal distribution* and analyses to test group means (when data is best represented by the mean). It can perform well with continuous **non-normal data** sample sizes below.

Parametric data test

- Data is normally distributed - mean and median are close together
- Interval or ratio data
- Variances between groups are similar

Non-Parametric data test

- Data is not normally distributed- the mean and median are VERY different
- Data is best represented by the median
- Nominal and ordinal data
- Small sample sizes
- Outliers can't be excluded

Inferential tests

Use this website for inferential tests: https://www.socscistatistics.com/tests/

T- tests (paired and unpaired)

Parametric analyses	Sample size guidelines for nonnormal data
1-sample t test	Greater than 20
2-sample t test	Each group should be greater than 15

- A *t* test is used to compare two means to reveal how significant the differences are
    - i.e., reading achievement of boys and girls
- There are several different types of t tests, depending on your data and experimental method
    - *Paired T test*: Test the same subject at two different times (repeated measures)
    - *Un-paired T tests*: Test different subjects (independent groups design)
    - *Two tailed paired/non-paired T tests* **both** greater and lower variability (e.g., scores could go up or down)
    - *One tailed paired/non-paired T tests* for **either** greater and lower variability (e.g., scores will go down)
- T tests provide a T score - the ratio between the difference between two groups and the difference within the groups.
- **The larger the t score, the more difference** there is between groups.
- **The smaller the t score, the more similarity** there is between groups.
- T-score has a p-value to go with it.
- A p-value is the probability that the results from your sample data occurred by chance assuming the null hypothesis were true

*Remember a null hypothesis is where the independent variable does not affect the dependent variable

The Mann-Whitney U test

It is a test of the null hypothesis. (Doesn't require a normal distribution). Typically used for ordinal data to test if two means are equal or not. Calculates a U statistic

- U statistics (like T scores have a P value to go with it)

The Wilcoxon signed-rank-test

It is an alternative to the t-test when the population cannot be assumed to be normally distributed. Used to compare 2 sets of scores from the same set of participants

- Wilcoxon sign test output is a Z value
- Z values have an associated P value

Measures of relationship- correlational design

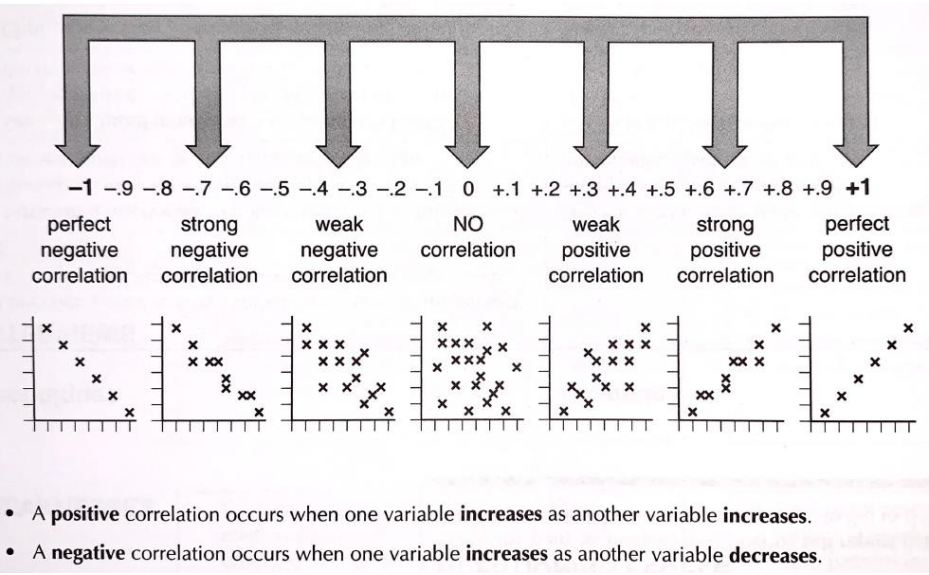
Correlation

It is a statistical measure of the strength and direction of the relationship between two variables (does not show a cause-and-effect relationship).

Scatter diagrams or scatter plots

It shows the relationship and correlations of variables.

Image 24: Scatterplots and correlation



Creating scatterplots excel

Creating scatterplots allow you to determine whether there is a relationship between 2 variables.

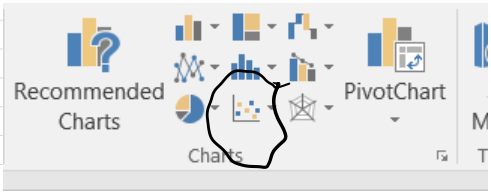
Follow these instructions to create scatterplots:

Insert your data into an excel spreadsheet (see figure below)

Highlight the 2 sets of data and click insert, and choose the scatterplot icon

Image 25: inserting scatterplots into excel

mean hours of sleep	stress score
7.12	4
5.38	5
7.15	4
6.39	3
7.69	4
6.04	4
8.32	4
6.18	5
8.05	5
7.37	3
7.24	4
6.39	4

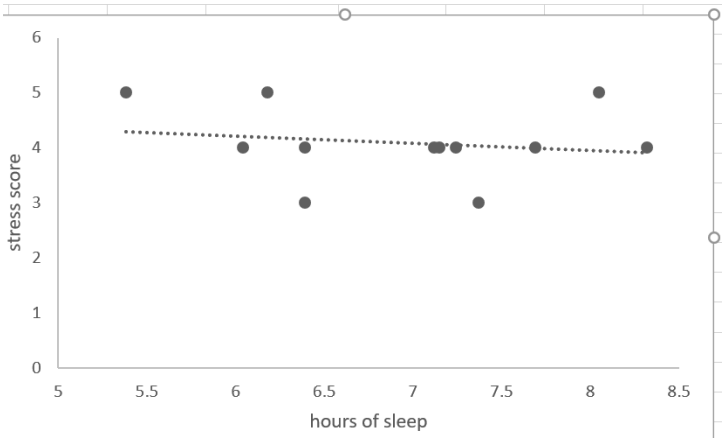


Remember to:

- add axis titles
- add a trendline
- remove gridlines
- adjust the axis to spread the data points out if needed
- change the colour to black

When your chart appears, you can use the + button at the side of the chart to edit it.

Figure 3: Correlation of hours of sleep and stress.

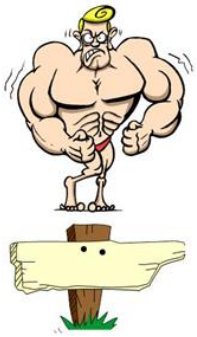


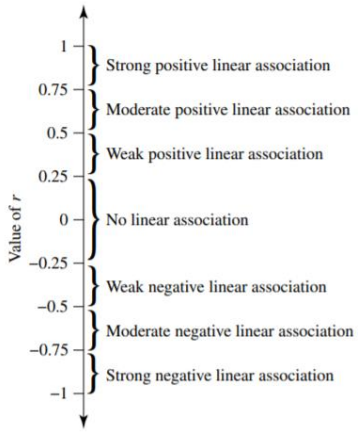
Pearson's Correlation coefficient (r)

It shows the strength of the relationship & is expressed as R score between -1.0 and +1.0. The positive & negative sign shows if it is a -ve or +ve relationship. The higher the number the stronger the correlation.

Image 26: Correlation coefficients

- **Correlation Coefficient**: number between +1 and -1 that represents the strength and direction of the relationship between two variables
- Correlations that are closer to +1 and –1 are stronger and are better able to accurately predict





Coefficient of Determination (R^2)

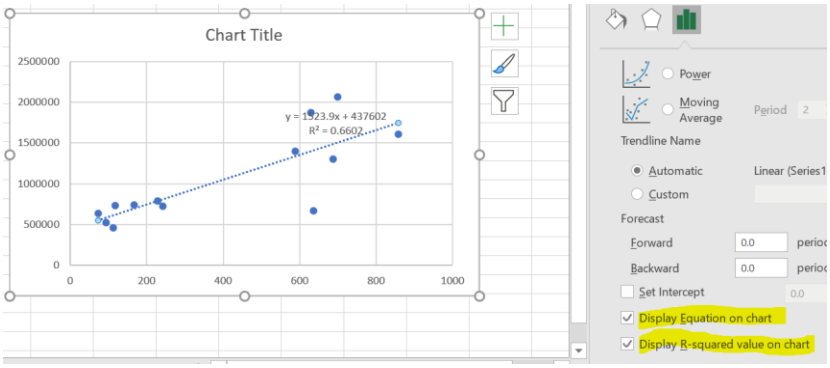
- It is the square of pearson's correlation coefficient
- Measured on a scale from 0 to 1
- It is a more accurate measure of the influence of the x-variable on the y-variable.

Interpreting R^2

For an r^2 value of 0.71, 71% of the variation in the y-variable is explained by the variation in the x-variable and 29% is explained by other factors.

Inserting an R^2 value in excel

Image 27: Inserting an R^2 value in excel



1. Click the + symbol and add a trendline
2. Click more options under the trendline and scroll to the bottom
3. Tick the Display Equation and R-squared vale on chart – highlighted in Image 27

Drawing conclusions and generalizations

A **conclusion** is a decision on what the results mean. This should be stated in terms of the original hypothesis. So, **a conclusion** is that **the hypothesis is rejected or supported**.

- Do not write – 'the hypothesis was proven/disproven'. (Another hypothesis might explain the relationship better).

A **generalisation** is about whether the research findings can be applied to the population represented by the sample. To generalise, the following conditions must be met:

- 1. The sample must represent the population
- 2. The results must be of statistical significance.
- 3. The effects of all confounding variables must be controlled.

Errors- type 1 and type 11

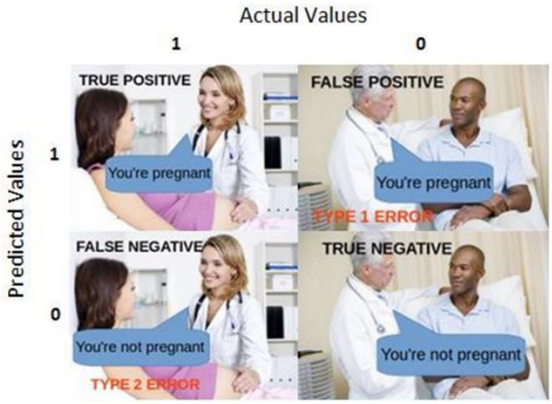
- A **type I error** when the null hypothesis is rejected BUT is in fact true
- A **type II error** has occurred if the null hypothesis is incorrectly kept.

Given the Null Hypothesis Is

Your Decision Based On a Random Sample	True	False
Reject	Type I Error	Correct Decision
Do Not Reject	Correct Decision	Type II Error

Two Types of Errors in Decision Making

Image 27: Errors in Decision making



Symbols/ equations

N= sample size

n= sample size

∑= sum of

$\bar{x}$= mean

μ = average population mean

r or R = Pearson's correlation co-efficient

R^2 = Coefficient of determination

t= t score

u= u score

p= probability score

$p \leq 0.05$ probability of the difference resulting from chance alone is less than 0.05 or there is a 99.95% chance that this difference is a result of the independent variable.

s= standard deviation or SD or $SD\ \bar{x}$ standard deviation of the sample

CI= confidence interval

SE = standard error of the mean

Data test preparation

Cognitive verbs:

The cognitive verbs in the test, direct you as to what skill you are to use. They can also help you determine the level of detail required.

The following verbs will be used. Ensure you are familiar with them:

Determine: work out and give an answer (usually one word answer/ number)

Calculate: work out and SHOW YOUR WORKINGS

Identify: analyse the evidence and make note of the correct answer (brief)

Contrast: compare to identify the relationship and explain similarities and/or differences (longer answer)

Distinguish: note the differences between data and scores and refer to the data in your answer e.g. (1.56)

Draw conclusions: based on analysis of the data (use a sentence)

Infer: draw a conclusion based on analysis (use a sentence)

Deduce: reach a conclusion based on analysis- give the answer and a reason (sentence)

Predict: give an expected result of an upcoming action or event; suggest what might happen based on available information.

Skills Checklist:

Ensure you can do the following things:

- ☐ Read a correlation scatterplot and understand the direction and strength. If the dots are close to the line, it is LINEAR.
- ☐ Understand what an R (r) score is and what it means. Understand what a positive and negative score mean and how to determine its strength and significance.
- ☐ Work out a mean, median and mode, using the **correct formulas** (see section on mean and mode above). **You must show working out.**
- ☐ Draw conclusions from evidence. What can be concluded from the results? What relationship or effect is evident?
- ☐ Standard deviation: what is it? what does it mean?
- ☐ P values: what are they and what is a significant p value
- ☐ Variability: how can you see variability in the descriptive stats?
- ☐ Inferential tests: how to choose which test to use. Reasons why you have chosen the test (at least 2 reasons)
- ☐ Error bars and confidence intervals: what are they? what do they mean? how are they different from each other? how do you interpret them? How do you use them to draw conclusions?

Work through the practice data test and answers in Oxford textbook.

Research Methods in Psychology

Research claim: A statement describing an idea that has yet to be proven as scientifically accurate
Example: Girls like Justin Bieber music more than boys.

Research question: a refined question to focus the research.
Example: Do teenage girls report a higher enjoyment of Justin Bieber music than teenage boys?

HYPOTHESES

Alternative hypothesis:
Predicts that there is a relationship between the IV and the DV.
Example: Teenage girls report a higher enjoyment for Justin Bieber music than teen boys.

Null hypothesis:
Predicts that there is no relationship between the IV and the DV.
Example: Teen girls and boys report an equal enjoyment of Justin Bieber music.

Variables

Independent variable (IV): a variable changed by the experimenter.
IV example: presence or absence of music.

Dependent variable (DV): a variable measured to test the effect of the IV.
DV example: length of time, correct number of answers.

Extraneous variables: variables that need to be controlled to avoid influencing the DV.
EV example: temperature, prior information.

Confounding variables: variables that will influence the DV.
CV example: additional time to study or prior knowledge.

Ethical Considerations

Explain why ethics need to be considered in psychological research. There must be no psychological or physical harm to participants.

Participant's rights

Confidentiality: participants must not be identified in any way.
Voluntary participation: the right to refuse participation.
Withdrawal rights: the right to leave the study at any time.
Informed consent: must be given information prior to doing the study.
Deception in research: only allowed if prior knowledge would confound results. Not allowed to cause distress to participant.
Debriefing: participants are told the results of the study.

PARTICIPANT SELECTION (SAMPLING THE POPULATION)

Random sampling: every member of the population has an equal chance of being selected.
Example: Gold lotto, census data.

Stratified sampling: ensuring the sample proportions matches the population.
Example: 30,000 people in the population but of those there are 24,000 males and 6,000 females. Sample could be 240 males and 60 females.

Convenience sampling: using what is available at the time.
Example: Psychology class.

DESCRIPTIVE RESEARCH

Correlational design: attempts to determine if there is a relationship between two variables.
Example: hours of technology use and grades in Science.

EXPERIMENTAL METHOD DESIGNS

Independent groups: equal participants are allocated to the two groups randomly.
Example: control group does maths problems, experimental group does it after coffee.

Matched participants: subjects are selected to match key characteristics in control and experiment groups.
Example: smoker & non-smoker groups.

Repeated measures: each participant is part of the control and the experiment groups. Counterbalancing is needed to avoid order effect.
Example: sleep with/out music.

Developing a Research Question from a Claim

Step 1: Identify the key words
Example: Social media changes relationships.
Step 2: Question key elements.
Example: E.g. what type of social media?
Step 3: Pose possible research questions.
Example: Does Facebook influence relationships?
Step 4: Critique the questions.
Example: Is it measureable? Is there 1 IV?
Step 5: Refine the question.
Example: Are teenagers less satisfied with their relationship with their parent/s if they use Facebook daily?

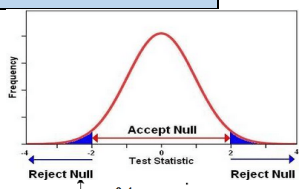
VALIDITY	Internal validity	Content validity	Construct validity	External validity
Definition	Is the result due to the IV?	Instrument measures what it says it does.	Does the test support the theory?	How natural is it? Ecological validity.
Example of low/high	Low = IQ test that just does maths	Sleep monitors.	IQ tests cover all intelligences	Sleep in a lab.

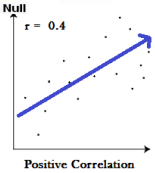
RELIABILITY	Internal reliability	Inter-rater reliability	Parallel form reliability	Test-retest reliability
Definition	All items in the test are equal.	Same scores with different experimenters	More than one form of measurement	Same results if repeated at a later time
Example of low/high	E.g. high scores for odd and even questions.	E.g. driving test despite assessor	Pre- and post-testing for a new drug.	Having the same score on a Psych test.

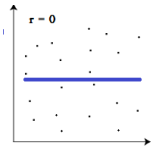
Data Analysis in Psychology

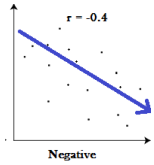
	TYPES OF DATA & DESCRIPTIVE STATISTICS			
	NOMINAL	**ORDINAL**	**INTERVAL**	**RATIO**
Definition	Data is qualitative, no ranking or order.	Data that has a sequence but irregular gaps between levels	Steps in the scale are evenly placed but zero does not mean zero.	Steps in the scale are evenly placed and zero means zero.
Example	Hair colour Religion	Ages of people Body mass	Temperature	Length Time
Measure of central tendency *(shows the central point of the data set)*	Mode	Median	Mean	Mean
Measure of dispersion *(shows the degree of difference between individual scores)*	NA	Inter-quartile range	Standard deviation	Standard deviation

	DESCRIPTIVE STATISTICS			INFERENTIAL STATISTICS	
Type of measurement	**Standard Deviation (SD)**	**Standard Error of the Mean (SEM)**	**Confidence intervals (CI)**	***p-values***	**r-values (correlation)**
Definition:	The dispersion of individual data points from the mean. Larger SDs mean a greater variation in the data.	Similar to SD, it uses the SD to find the variation from the mean. A large SEM means more variability from the mean.	A CI is 2x SD from the mean. It shows where 95% of the data falls within that range.	*p-value* refers to the probability that any difference is due to chance. It indicates if there's a **statistically significant difference** between two conditions.	Correlation coefficients (**r**) describe the **relationship** between two variables (e.g. IQ and hair colour).
Use/Examples:	A SD of 0.32 means a greater dispersion/ variation of data points than an SD of 0.12.	A greater SEM indicates more variation between data points.	If two CIs do not overlap, they are **likely** to be statistically different.	IF $p \leq 0.05$ THEN you can accept that a difference is **statistically significant**. IF $p > 0.05$ THEN the conditions are **not** statistically significant.	r = +1 = positive correlation r = 0 = no correlation r = -1 = negative correlation









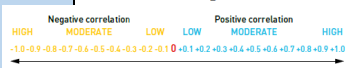


FIGURE 1.26 Negative and positive correlation

	STATISTICAL ANALYSIS – CHOOSING INFERENTIAL STATISTICS		
	CORRELATION	**INDEPENDENT GROUPS**	**REPEATED MEASURES OR MATCHED PARTICIPANTS**
PARAMETRIC DATA Use if: - Normally distributed - Large sample size - Interval/ratio data	**P**earson's correlation	**U**npaired t-test	**P**aired t-test
NON-PARAMETRIC TESTS Use if: - Nominal/ordinal data - Distribution unclear (skewed/outliers) - Small sample size (>20) - Less statistical power	**S**pearman correlation	**M**ann-Whitney U test	**W**ilcoxon signed-ranks test

	TYPE I ERROR – False Positive	TYPE II ERROR – False Negative
Definition:	Finding a "significant effect" ($p < 0.05$) when there is no true effect. So saying something is wrong when it is in fact right.	Failing to find a "significant effect" ($p < 0.05$) when there is a true effect.
Examples:	Pregnancy test reporting a man he is pregnant	Pregnancy test reporting a pregnant woman isn't pregnant